

# Regression with non-linear terms

## Beyond the linear Bayesian model

Garyfallos Konstantinoudis

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### A clarification

In the name normal (or Poisson, binomial, etc.) linear model, the word ‘linear’ refers to the response being modelled as a linear combination of covariates, i.e.

$$Y_i \sim N(\beta_1 + \beta_2 X_{i2} + \dots + \beta_p X_{ip}, \sigma^2)$$

It does not refer to each covariate-response relationship being linear. Therefore the following is also within the class of normal linear models

$$Y_i \sim N(\beta_1 + \beta_2 X_i + \beta_3 X_i^2, \sigma^2)$$

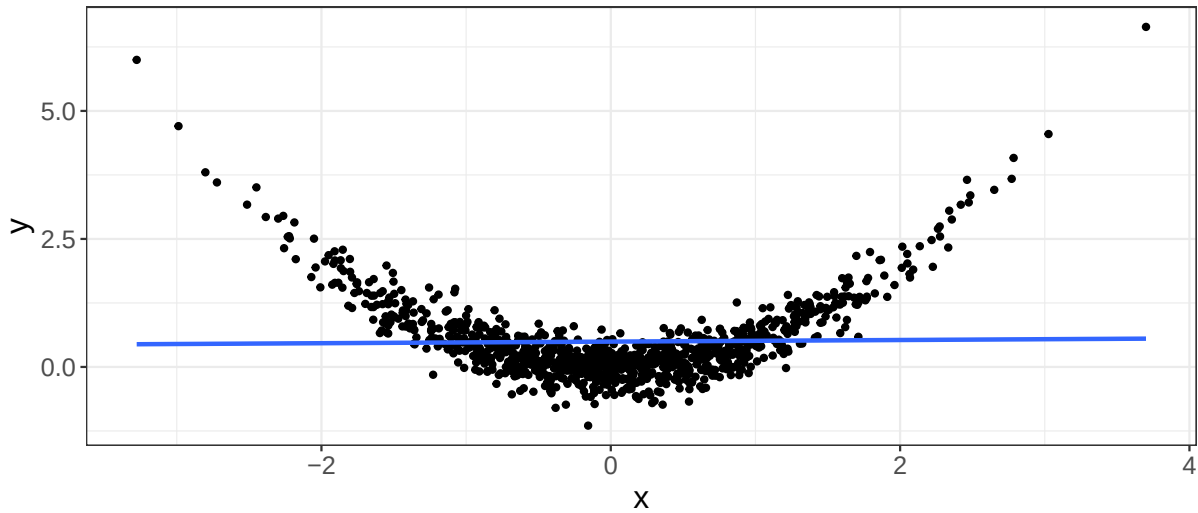
Here, the relationship between Y and x is quadratic, but it is still a linear model.

### So how do you tell if the relationship is linear?

- Plot each covariate against the response and see what shape the relationship is.
- Make sure you plot it with the transformation used in the model.
- If linear, fit a linear model, if not start thinking of the shape of the relationship.
- Fit as simple a model as possible, do not overcomplicate it unnecessarily. (Occam’s razor)

## Example

Associations are not solely linear

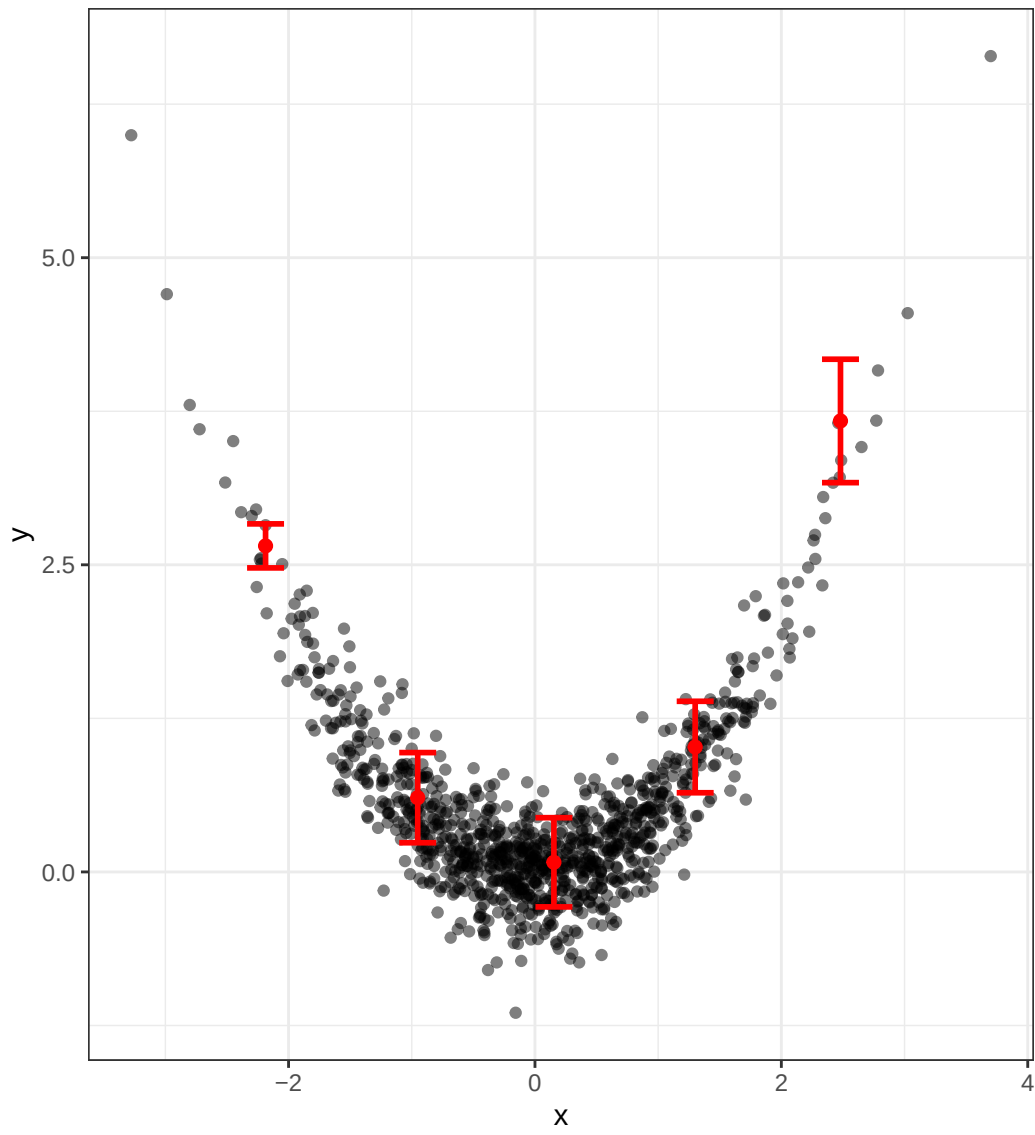


## Ways to model non-linearity

- Transform the continuous exposure to categorical variable and model the categories.
- Basis functions (e.g., polynomial, Fourier)
- Natural splines
- Gaussian priors (e.g., random walks, kernels)

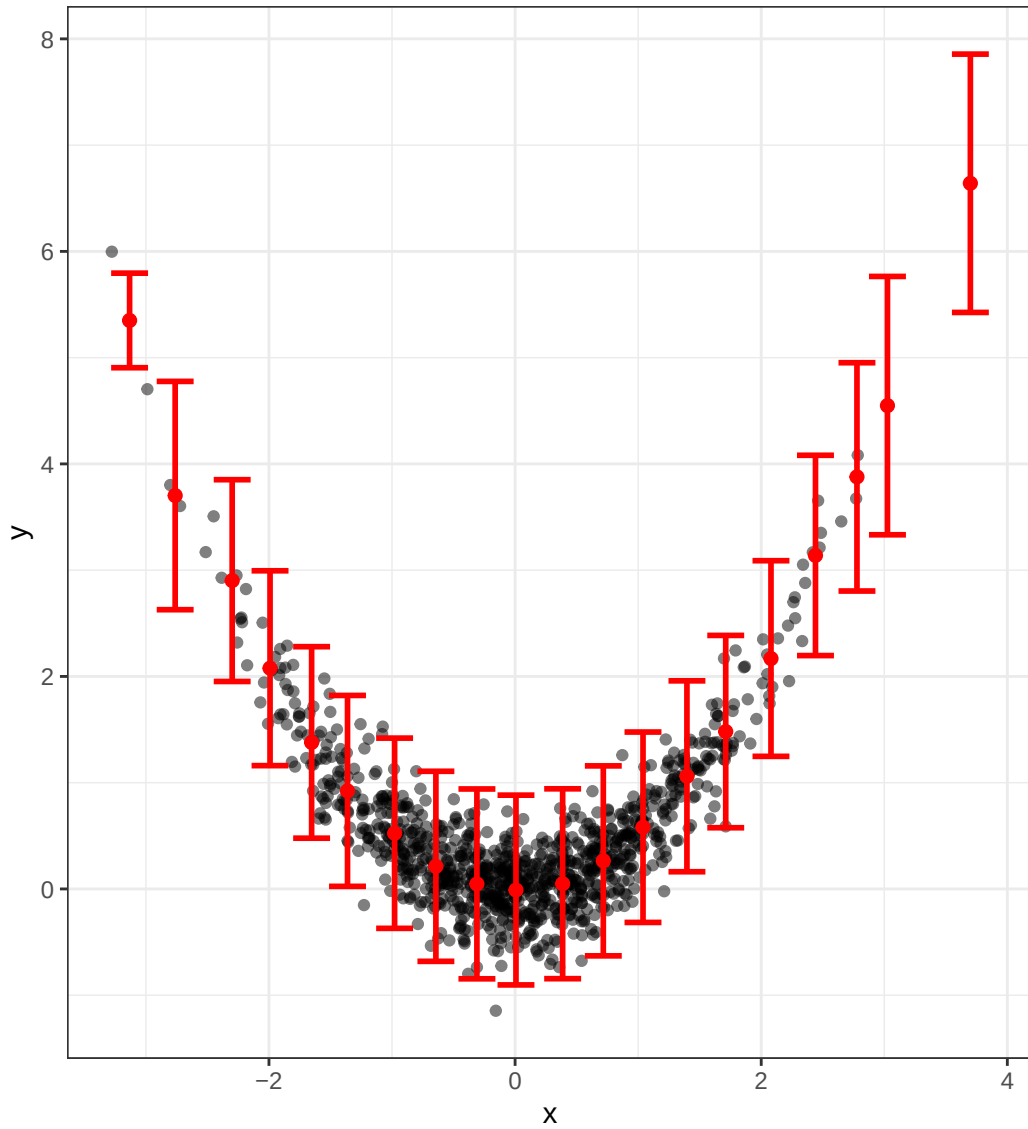
## Categorical transformation

- Let's first consider quintiles
- What's the problem with this fit?



### Categorical transformation

- More categories...
- What's the problem with this fit?



## Basis Expansions

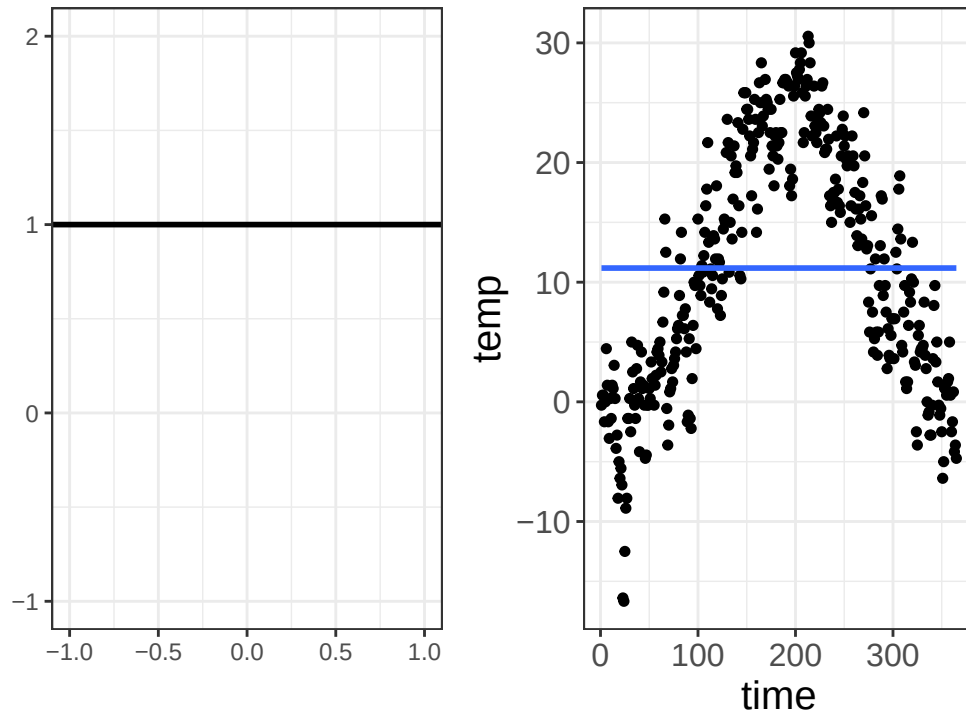
We need to define a set of flexible functions that could capture relationships that are not linear. In general, we can write:

$$Y_i = \sum_j^K \beta_j \phi_j(X_j) + \epsilon_i$$

which we can write:  $f(X) = \beta^T \Phi(X)$  and we say that  $\Phi(X)$  is a basis system for  $f$ .

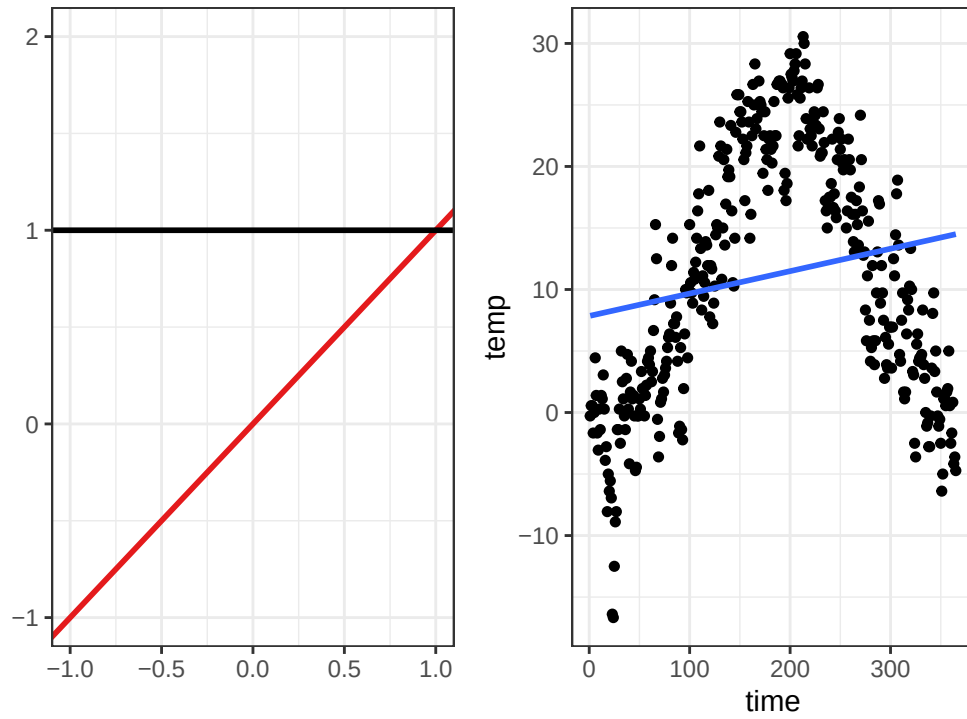
### The polynomial basis function

$$\Phi(X) = (1) \text{ ,thus } Y_i = \beta_0 + \epsilon_i$$



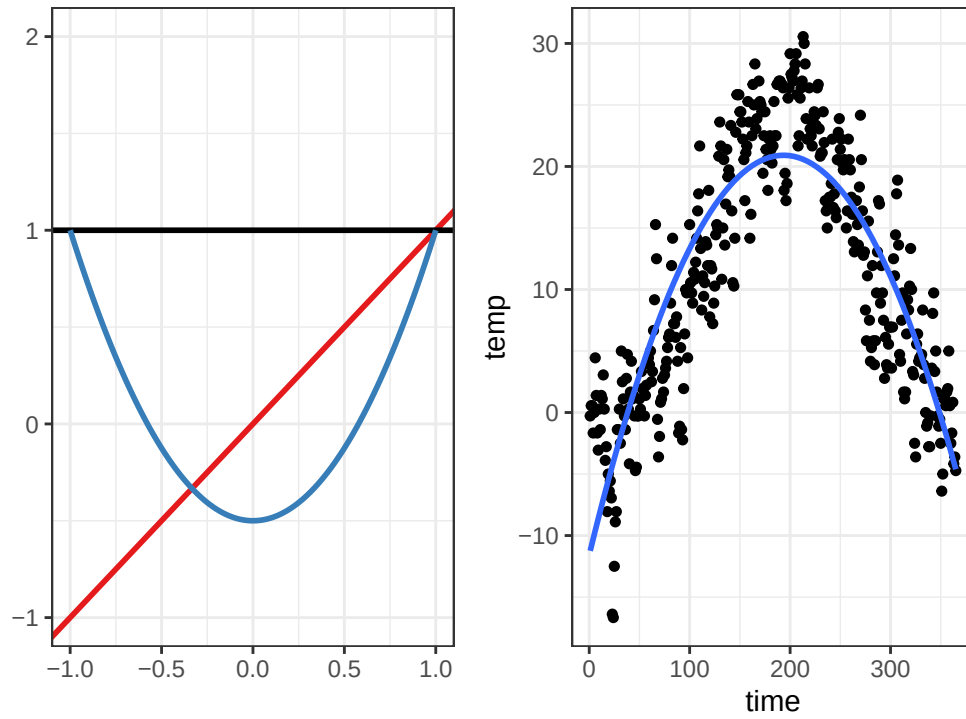
### The polynomial basis function

$$\Phi(X) = (1X) \text{ ,thus } Y_i = \beta_0 + \beta_1 X_1 + \epsilon_i$$



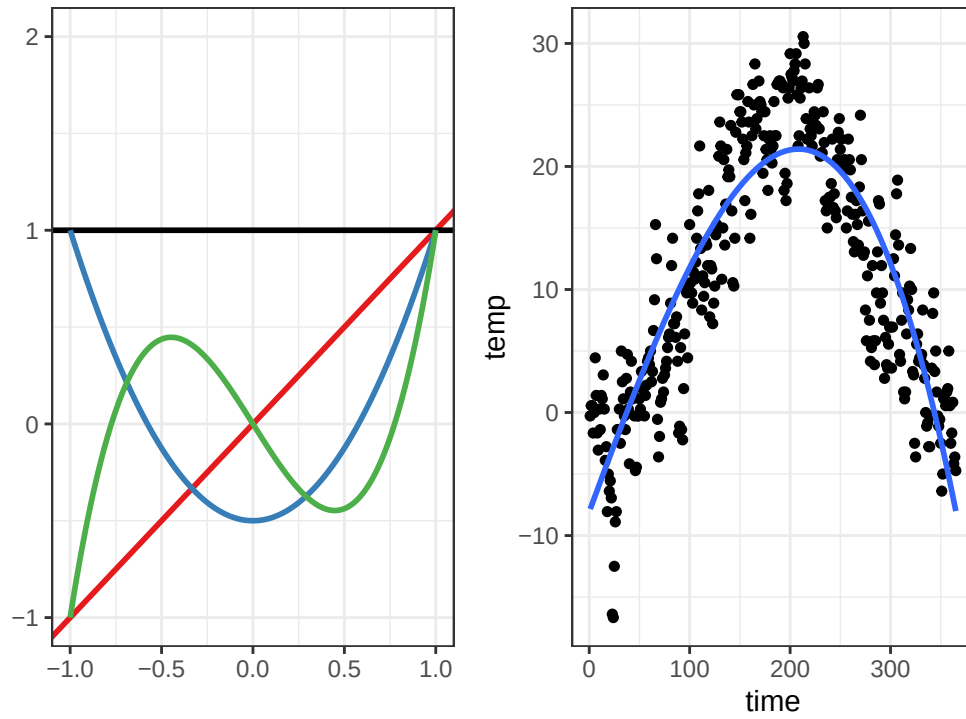
**The polynomial basis function**

$$\Phi(X) = (1 \ X \ X^2) , \text{thus } Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon_i$$



**The polynomial basis function**

$$\Phi(X) = (1 \ X \ X^2 \ X^3) , \text{thus } Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon_i$$

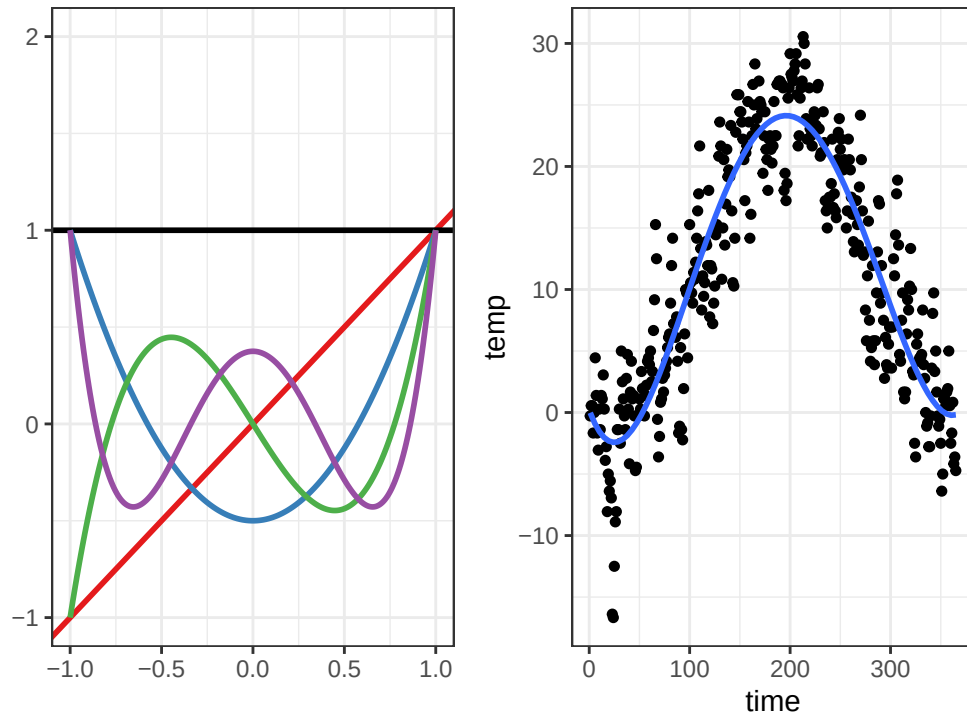


### The polynomial basis function

$$\Phi(X) = (1 \ X \ X^2 \ X^3 \ X^4), \text{thus}$$

$$Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \epsilon_i$$

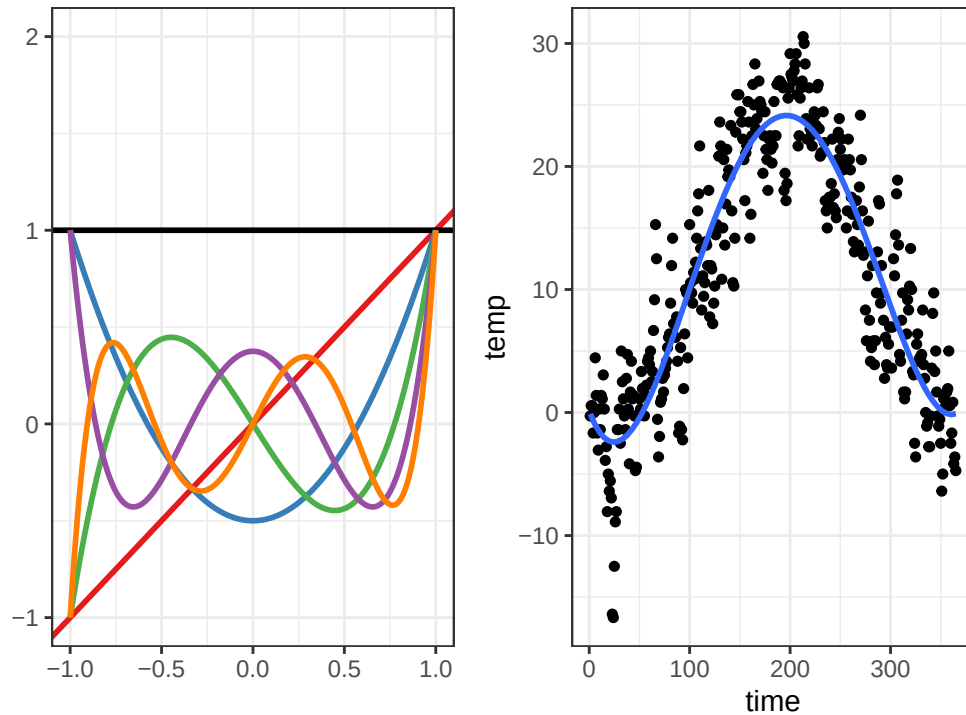




### The polynomial basis function

$$\Phi(X) = (1 \ X \ X^2 \ X^3 \ X^4 \ X^5), \text{ thus}$$

$$Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5 + \epsilon_i$$



## Pros and cons

### Pros

- These curves are quite flexible—a quadratic can fit most biologically plausible curves
- The curves only use 1(quadratic) or 2(cubic) degrees of freedom more than linear, unlike dummy variable models
- The results are not sensitive to choice of boundaries (there aren't any)
- Outliers mostly influence the extremes of the curve, not the center part

### Cons

- They use more degrees of freedom than linear, and therefore have less power
- There is still some sensitivity to influential observations

## Other types of basis function

- **Harmonics** - The Fourier basis function

$$1, \sin(\omega X), \cos(\omega X), \sin(2\omega X), \cos(2\omega X), \dots,$$

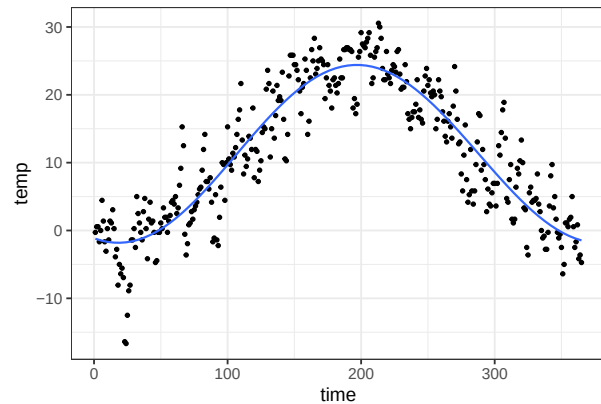
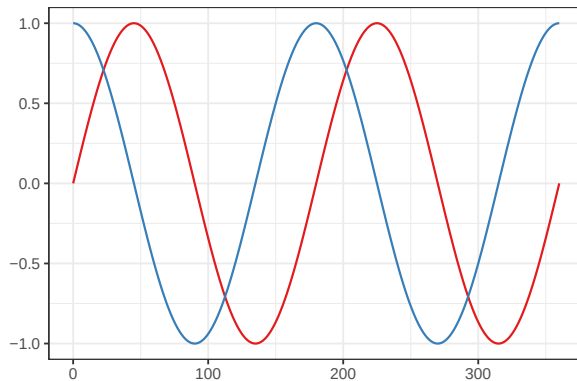
$$\sin(m\omega X), \cos(m\omega X)$$

- constant  $\omega$  defines the period of oscillation of the first sine/cosine pair. This is  $\omega = 2\pi/P$  where  $P$  is the period.

## Example: Fourier basis function

$$\Phi(X) = (1 \sin(\omega X) \cos(\omega X) \sin(2\omega X) \cos(2\omega X)) \text{ ,thus}$$

$$Y_i = \beta_0 + \beta_1 \sin(\omega X) + \beta_2 \cos(\omega X) + \beta_3 \sin(2\omega X) + \beta_4 \cos(2\omega X) + \epsilon_i$$



## Pros and cons

### Pros

- Excellent computational properties, especially if the observations are equally spaced.
- Natural for describing periodic data, such as the annual weather cycle
- The results are not sensitive to choice of boundaries (there aren't any)

### Cons

- functions are periodic; this can be a problem if the data are, for example, growth curves.

## Parametric non-linear effects

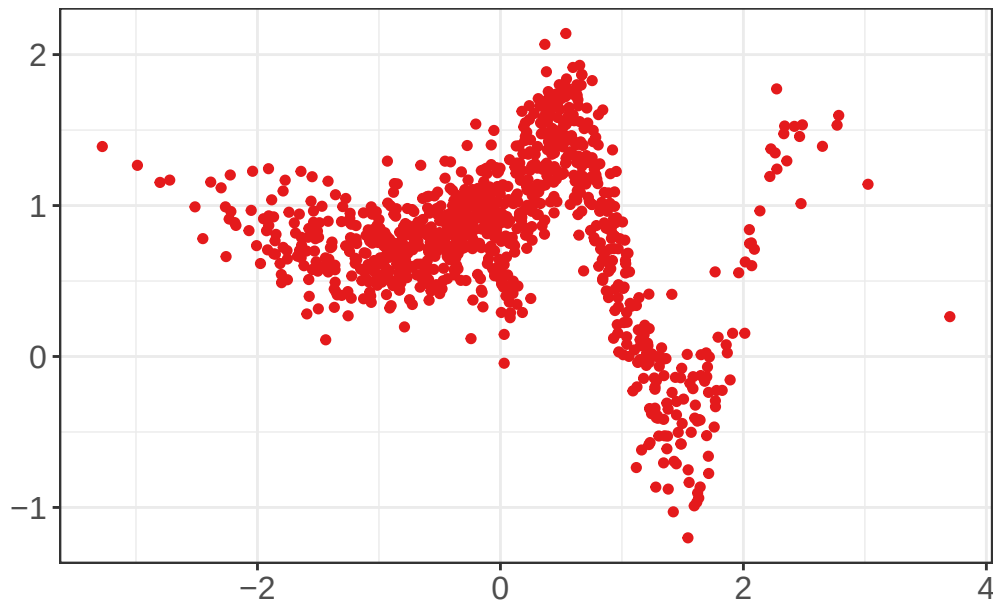
- The following models captured the non-linear relationships using simple non-linear functions.
- These types of models are to be preferred if possible, because again they are simpler to make inference from, e.g. the relationship is quadratic.
- However, there may be times when the relationship being modelled does not look like a parametric function. Then what should you do?

## Parametric non-linear effects

Consider the following form

```
set.seed(11)
x <- rnorm(n = 1000)
y <- numeric(1000)
y[x < 0] <- 1 + 0.5 * x[x < 0] + 0.2 * x[x < 0]^2 + rnorm(n = length(x[x < 0]), sd = 0.2)
y[x >= 0] <- 0.5 + sin(2 * pi / 2 * x[x >= 0]) + rnorm(n = length(x[x > 0]), sd = 0.3)

ggplot() +
  geom_point(aes(x = x, y = y), cex = 1.3, col = cols[1]) +
  theme_bw() +
  xlab("") +
  ylab("") +
  theme(text = element_text(size = 15))
```



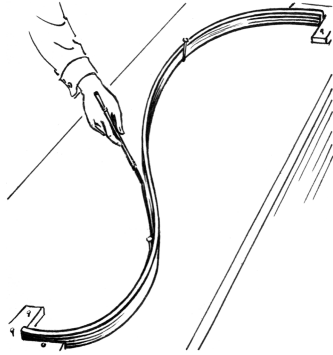
It doesn't look like any simple parametric form (e.g. polynomial, sinusoidal, etc), so what do you do?

### Smooth functions

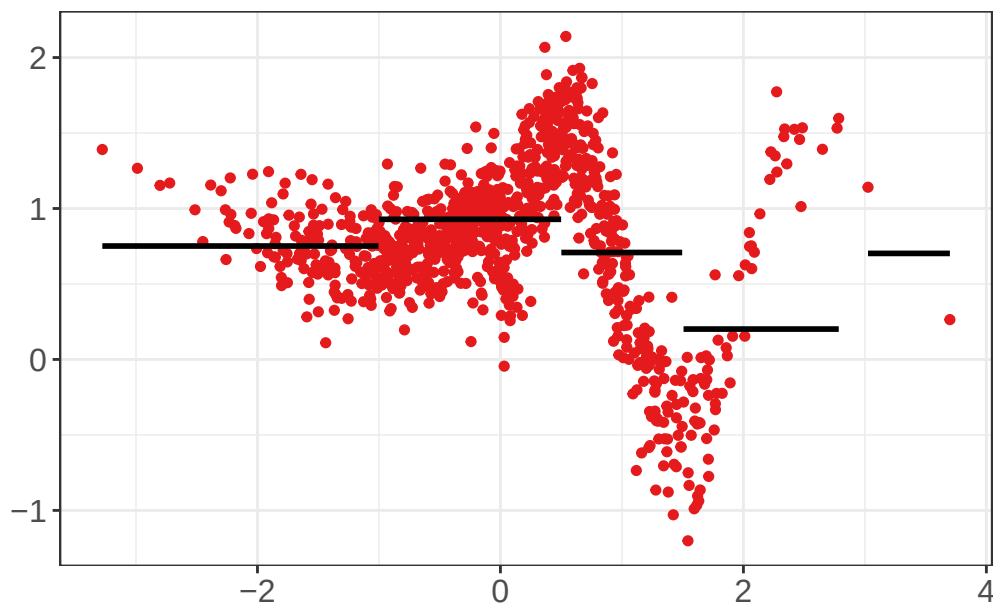
- There are many methods to estimate non-linear relationships such as that on the previous slide.
- They are generically called smooth functions, and include splines (lots of different types), kernel smoothers and local linear smoothers.
- We will focus on splines in this lab, because they are simple to understand graphically and are easy to fit to the data.

### Splines

- Splines were originally thin splints of wood used to trace complex, smooth curves in engineering and architectural.
- Splines were pinned to the drawing at points where the spline changed its curvature.



### Piecewise constant (splines?)



### Piecewise Linear Splines

- We begin fitting splines by dividing the range of exposure into pieces.
- Instead of fitting a constant in each piece, we fit a separate linear term in each piece.
- This has more power since it allows variation within categories to predict variation in outcome.
- We can use fewer categories to capture the deviation from a strictly linear curve because we have slopes within category.

## Linear splines

Linear threshold model

$$Y_i = \beta_0^{(1)} + \beta_1^{(1)}x + \epsilon_i \quad x \leq -1 \quad (1)$$

$$Y_i = \beta_0^{(2)} + \beta_1^{(2)}x + \epsilon_i \quad -1 < x \leq 0.5 \quad (2)$$

$$Y_i = \beta_0^{(3)} + \beta_1^{(3)}x + \epsilon_i \quad 0.5 < x \leq 1.5 \quad (3)$$

$$Y_i = \beta_0^{(4)} + \beta_1^{(4)}x + \epsilon_i \quad 1.5 < x \leq 3 \quad (4)$$

$$Y_i = \beta_0^{(5)} + \beta_1^{(5)}x + \epsilon_i \quad x \geq 3 \quad (5)$$

(6)

Linear spline model

$$Y_i = \beta_0 + \beta_1x + \beta_2(x+1)_+ + \beta_3(x-0.5)_+ + \beta_4(x-1.5)_+ + \beta_5(x-3)_+ + \epsilon_i$$

$$(x-k)_+ = \begin{cases} 0, & x < k \\ x-k, & x \geq k \end{cases}$$

## Linear splines in R

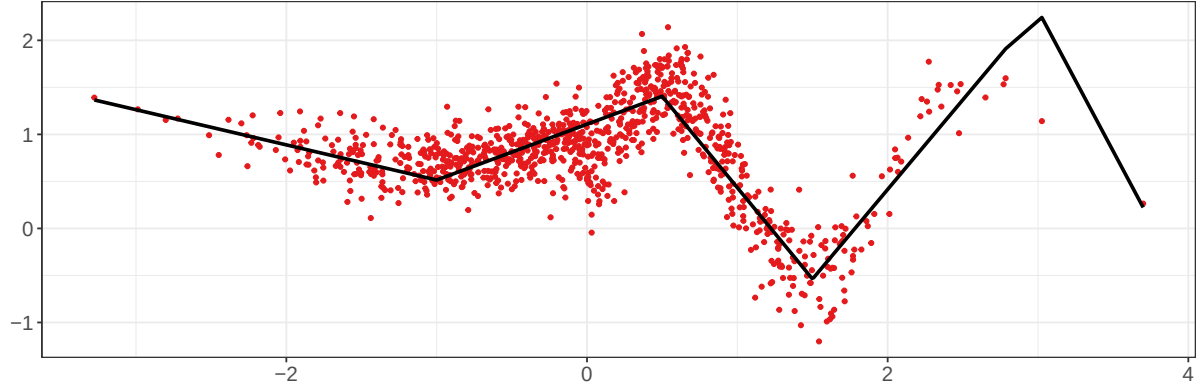
- You can use the `bs()` function in R to define the basis function:

```
bs(dat$x, degree = 1, knots = c(-1, 0.5, 1.5, 3)) %>%  
  head()
```

|      | 1         | 2          | 3          | 4 | 5 |
|------|-----------|------------|------------|---|---|
| [1,] | 0.7273541 | 0.27264593 | 0.00000000 | 0 | 0 |
| [2,] | 0.3156038 | 0.68439625 | 0.00000000 | 0 | 0 |
| [3,] | 0.7731419 | 0.00000000 | 0.00000000 | 0 | 0 |
| [4,] | 0.8407311 | 0.00000000 | 0.00000000 | 0 | 0 |
| [5,] | 0.0000000 | 0.32151084 | 0.6784892  | 0 | 0 |
| [6,] | 0.9561009 | 0.04389912 | 0.00000000 | 0 | 0 |

## Linear splines in R

- And plot the result:



## Cubic splines

Similarly we can define higher order polynomial splines, for instance:

$$Y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \quad (7)$$

$$\beta_4 (X + 1)_+ + \beta_5 (X + 1)_+^2 + \beta_6 (X + 1)_+^3 + \quad (8)$$

$$\beta_7 (x - 0.5)_+ + \beta_8 (x - 0.5)_+^2 + \beta_9 (x - 0.5)_+^3 + \quad (9)$$

$$\dots + \epsilon_i \quad (10)$$

$$(x - k)_+ = \begin{cases} 0, & x < k \\ x - k, & x \geq k \end{cases} \quad (11)$$

which reduces to the following to ensure smooth curvature on the knots (it can be seen after deriving the first and second derivative):

$$Y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \quad (12)$$

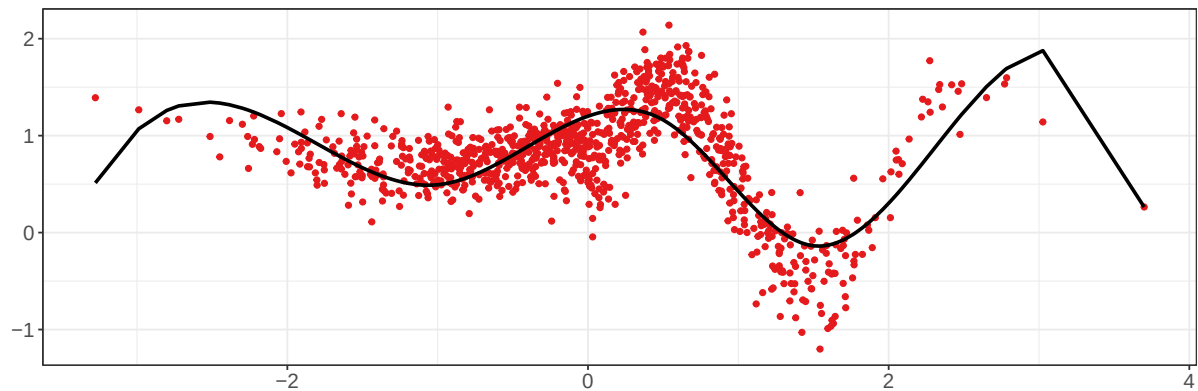
$$\beta_6 (x + 1)_+^3 + \beta_9 (x - 0.5)_+^3 + \dots + \epsilon_i \quad (13)$$



## Cubic splines

- Similar as before:
- `bs(x, degree = 3, knots = c(-1, 0.5, 1.5, 3))`:

without the splines package:



## Splines

A spline of order  $n$  is a piecewise polynomial function of degree  $n - 1$  in a variable  $x$ .

(Basis) Splines can be:

- Piecewise constant.
- Linear.
- Quadratic.
- Cubic.
- higher order polynomials.
- etc.

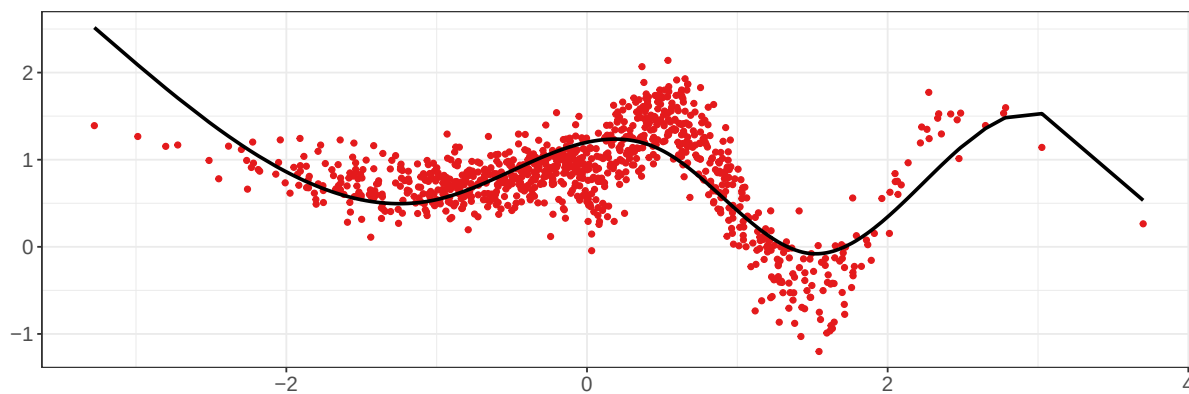
Why are the borders so wiggly?

## Natural cubic splines

- Splines can have high variance in the boundaries.
- This problem aggravates if knots too few.
- The natural spline constraints the fit to be linear before and after the first and last knots.
- This stabilises the fitting and makes a more reasonable assumption.

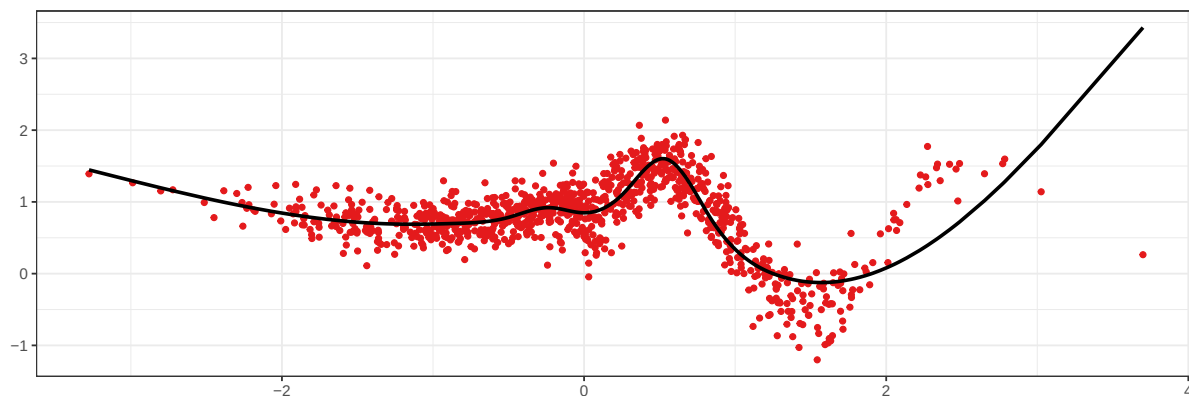
## Natural cubic splines in R

- To define natural cubic splines in R:
- `ns(x, knots = c(-1, 0.5, 1.5, 3))`:

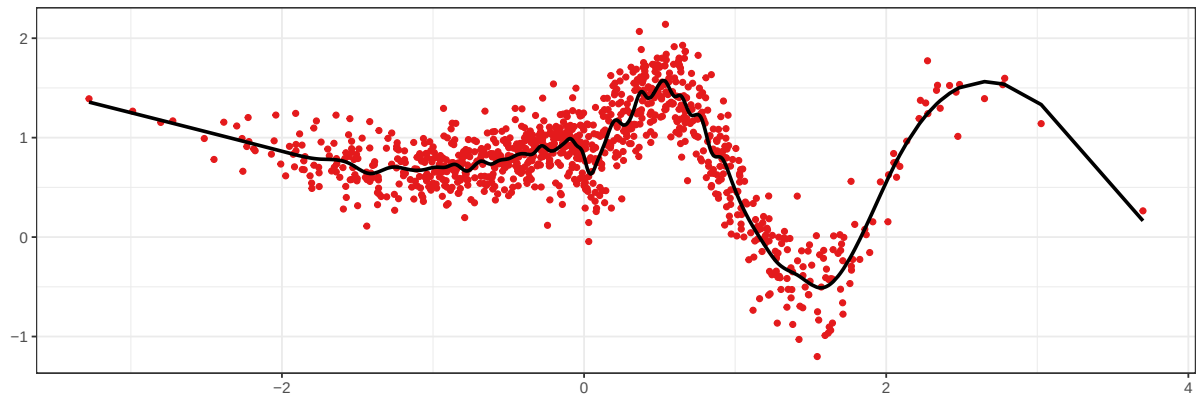


## Well, better fit be achieved with more knots

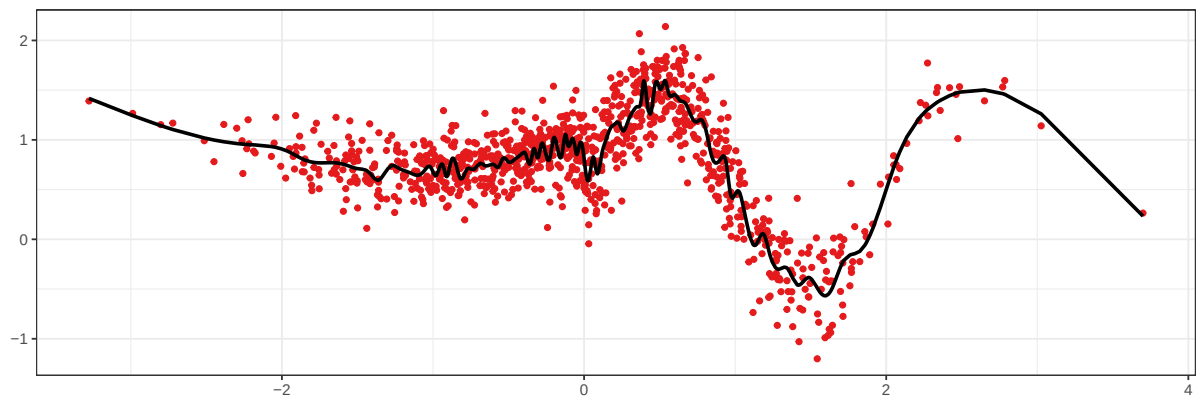
- `ns(x, df = 10)`



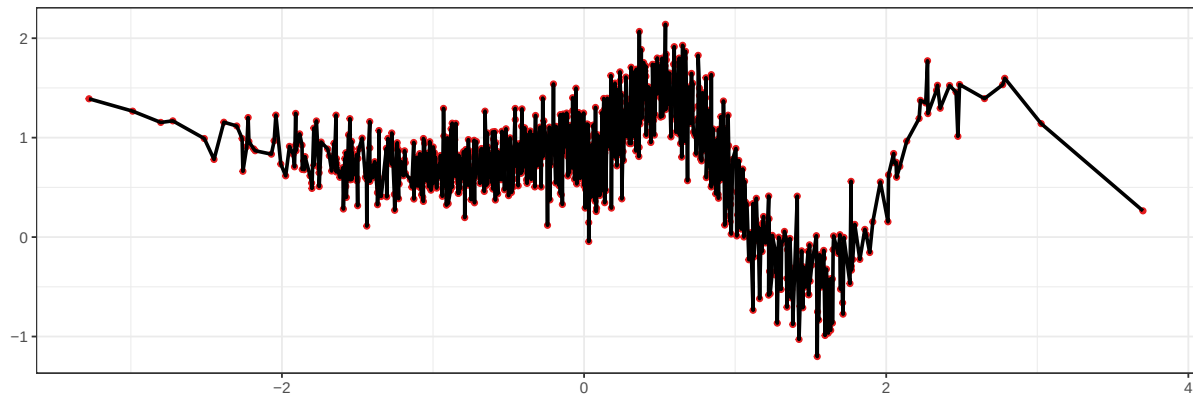
... and more



... and more



... and more



is this a useful model?

## Gaussian priors

- Random walks
- Autoregressive processes
- Gaussian kernels (out of the scope)
  - Correlation is a function of distance
  - “Continuous domain models”
  - Exponential, Gaussian, Matern, etc. (correlation functions)

## Random walk of order n

- Random walk of order 1

$$x_i = x_{i-1} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

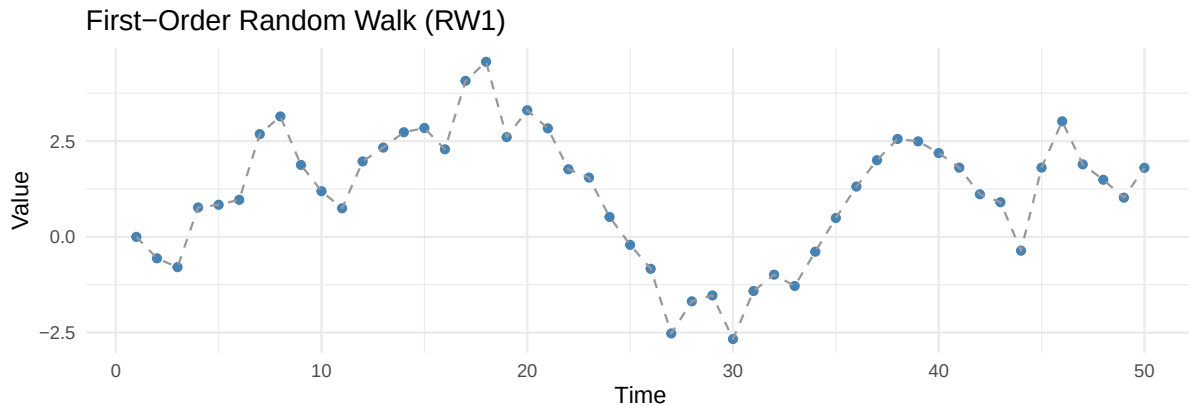
- Random walk of order 2

$$x_i = 2x_{i-1} - x_{i-2} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

- This is very similar with the categorical transformation example, but more concise.

The prior controls smoothness: the stronger it is the smoother the result. (given the data)

## Random walk of order 1 example



## Random walk of order 2 example

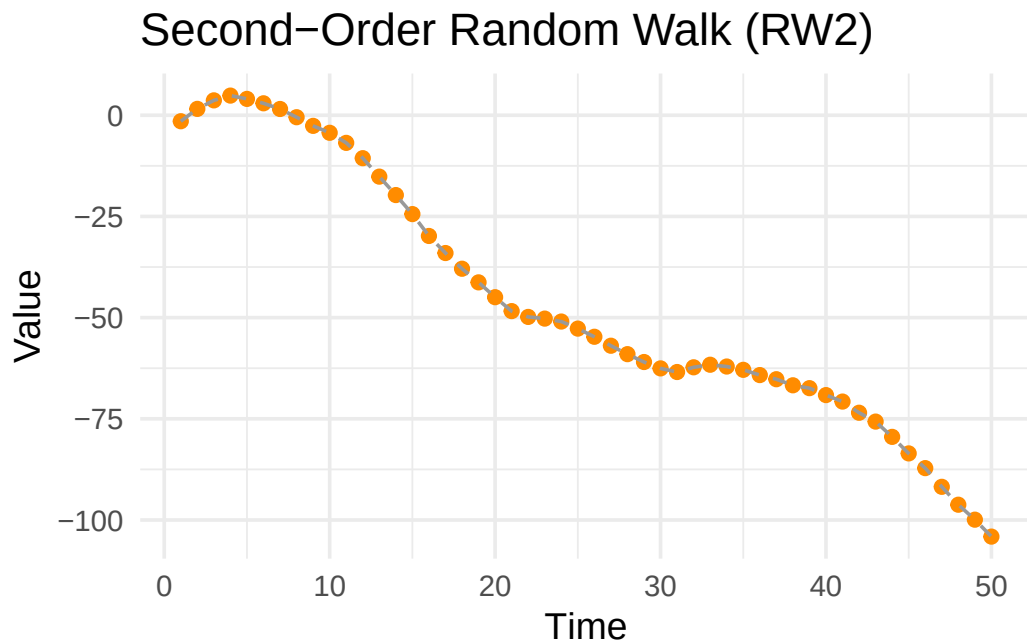
```
# Simulate RW2
set.seed(12)
n <- 50
sigma <- 1
x <- numeric(n)
x[1:2] <- rnorm(2, mean = 0, sd = sigma) # Initial values

for (t in 3:n) {
  x[t] <- 2 * x[t - 1] - x[t - 2] + rnorm(1, mean = 0, sd = sigma)
}

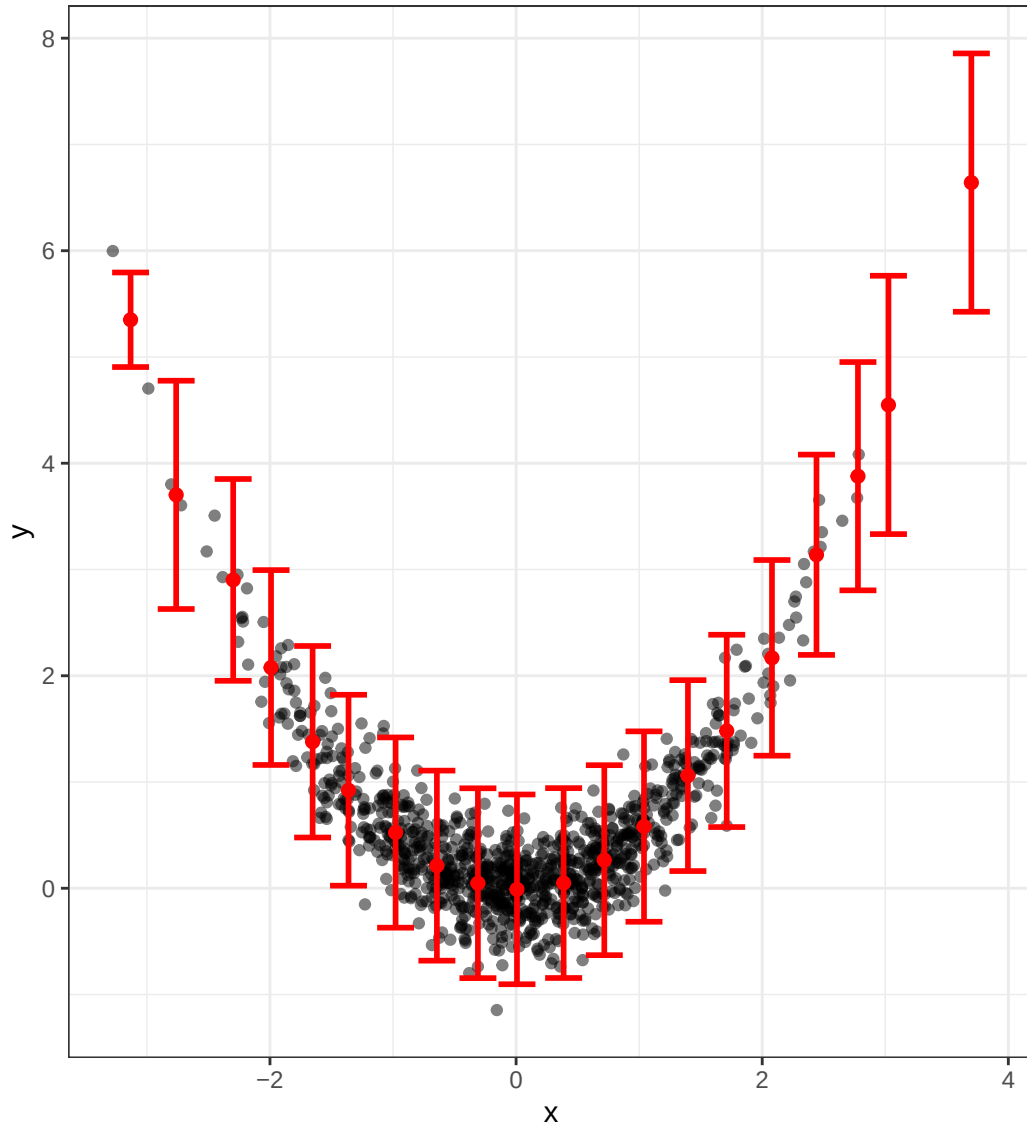
# Create data frame
df <- data.frame(
  time = 1:n,
  value = x
)

# Plot with ggplot2
ggplot(df, aes(x = time, y = value)) +
```

```
geom_point(color = "darkorange", size = 2) +
geom_line(color = "grey60", linetype = "dashed") +
labs(
  title = "Second-Order Random Walk (RW2)",
  x = "Time",
  y = "Value"
) +
theme_minimal(base_size = 14)
```



## Back to categorical

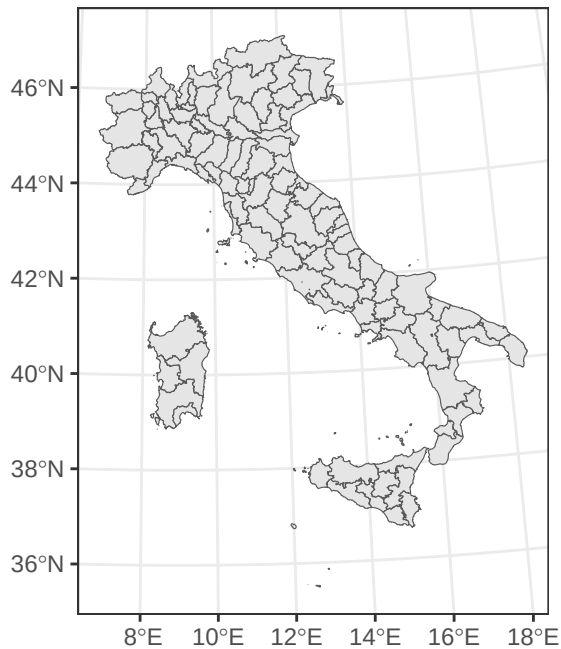


### Example: Mortality and temperature in Italy

- 1,946,755 summer deaths during 2011-2018 in Italy
- Data is available at 107 administrative units
- Temperature is available from ERA-5 at 9km resolution

Warning: package 'sf' was built under R version 4.6.1

Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf\_use\_s2() is TRUE



## Example

- Fit a linear, a categorical and a random walk model for capturing the effect of temperature.

Warning: package 'INLA' was built under R version 4.6.1

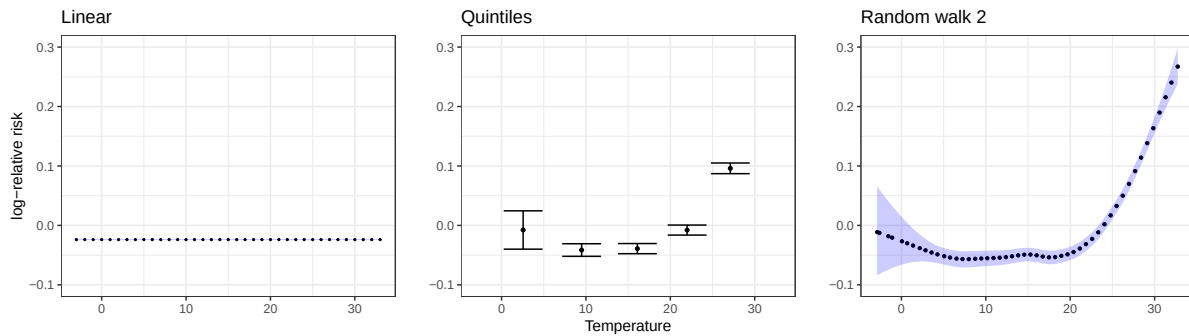
Loading required package: Matrix

```
Warning in `[<-.data.frame`(`*tmp*`, , y_vars, value = list(y =  
c(-0.0238641295256351, : replacement element 1 has 1369 rows to replace 37 rows
```

```
Warning in `[<-.data.frame`(`*tmp*`, , y_vars, value = list(ymin =  
c(-0.0229203899127436, : replacement element 1 has 1369 rows to replace 37 rows
```

```
Warning in `[<-.data.frame`(`*tmp*`, , y_vars, value = list(ymin =  
c(-0.0229203899127436, : replacement element 2 has 1369 rows to replace 37 rows
```





## Autoregressive process of order n

- Autoregressive process of order 1:

$$x_i = \phi x_{i-1} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

- Autoregressive process of order 2:

$$x_i = \phi_1 x_{i-1} + \phi_2 x_{i-2} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

- when  $\phi = 0$  we have RW1 or RW2
- To complete the parametrisation, we need to define priors of  $\phi$ . Can you think of a prior?

## Extensions

- It is straight forward to include confounders in the linear predictor.
- Confounders can be included as linear or more flexible terms.
- We can account for spatiotemporal correlations.
- We can incorporate the delayed effect of temperature on health (lags)
  - joint lag models
  - distributed non-linear lag models

## Summary

- Introduction to non-linear exposure models
- Theory and application of linear splines
- Understand more flexible non-parametric functions (focus on splines)
- Random walk of order 2 to model non-linearity

## Questions?